Fiscal Year:	FY 2012	**Task Last Updated:**	FY 02/19/2013
PI Name:	Berkowitz, Dan E. M.D.		
Project Title:	Radiation, Endothelial Cell Senescence, Accelerated Aging, and Atherosclerosis		
Division Name:	Human Research		
Program/Discipline:	NSBRI		
Program/Discipline--Element/Subdiscipline:	NSBRI--Cardiovascular Alterations Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	21287-8711	**Congressional District:**	7
Comments:			
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No. of Post Docs:	3	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
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COI Name (Institution):	Nyhan, Daniel (The Johns Hopkins University) Romer, Lewis (Johns Hopkins Medical School) DeWeese, Theodore (Johns Hopkins Medical School)		
Grant/Contract No.:	NCC 9-58-CA01301		
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Task Description:	NOTE: Dr. Dan Berkowitz took over as Principal Investigator (PI) sometime in 2012; see Shoukas for previous reporting (Ed., 2/12/2013) Exposure to ionizing radiation represents one of the most significant health risks to astronauts during both short, but particularly long, duration spaceflight. Vascular endothelial cells (ECs), the most radiosensitive vascular cell type, remain the major regulating interface between the blood and vessel wall. There, they act as a paracrine organ releasing signaling molecules that regulate vessels tone, permeability and cell and platelet adhesion. Dysfunction of the endothelium may result from both an increase in endothelial cell damage (oxidative stress with nitroso-redox imbalance), an impairment of the endothelial repair mechanisms (decrease in the number and function of endothelial progenitor cells) or an acceleration of endothelial cell senescence. Radiation is known to induce cellular oxidative stress, as well as suppress bone marrow proliferation. Data from our lab suggests that gamma and heavy iron ion radiation results in the activation of the ROS producing enzyme xanthine oxidase (XO). Activation of enzymes such as arginase also results in a decrease in endothelial protective nitric oxide (NO) with altered nitroso-redox balance and endothelial damage. We hypothesize that radiation will result in activation of ROS producing enzymes that will result in increased oxidative stress, nitroso-redox imbalance and endothelial cell damage. We tested the countermeasure hypothesis that the XO inhibitor (oxypurinol) will reduce radiation related endothelial dysfunction, We also tested the hypothesis that radiation alters endothelial cell repair as measured in an angiogenesis assay in vitro. Additionally, we tested the hypothesis that pharmacologic activation of the endogenous anti-oxidant master transcription factor, Nrf2 would attenuate oxidant injury in the vasculature. Finally, we tested whether radiation could alter coronary endothelial function in the same manner as other systemic arteries, given that the coronary circulation is the most critical with regard to translational relevance in the astronauts. We have demonstrated that radiation-induced endothelial dysfunction is largely due to the activation of the ROS producing enzyme, xanthine oxidase in rat aorta as reactive oxygen species derived from xanthine oxidase contribute to impaired endothelial nitric oxide production and attenuated endothelial-dependent vasorelaxation. We demonstrated that XO inhibition leads to a restoration of endothelial cell repair as measured by the same angiogenesis assay. We have developed an in vivo assay of coronary endothelial function using high resolution Doppler and both pharmacologic and physiologic activation of the endothelium. We have shown that radiation leads to an impairment of coronary endothelial function which is restored by acute administration of ROS inhibitors as well as the chronic activation of the endogenous antioxidant defense mechanism mediated by Nrf2. We are now exploring novel sensitive functional proteomic biomarkers of oxidant injury mediated by radiation.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	While it is well established that ionizing radiation has significant deleterious effects of cardiac and vascular structure and function, the molecular mechanisms contributing to this pathophysiology have been incompletely studied and understood. Epidemiologic data supports the thesis that radiation leads to an increase in both coronary, myocardial, and valvular heart disease. It is also well established that the most important early event leading to atherosclerosis and vascular aging and senescence is endothelial dysfunction and damage. During the course of this proposal we have 1) Established the role and lower limit of dose at which radiation leads to and increase in oxidative stress and alteration in nitrso-redox balance which contribute to endothelial dysfunction. 2) We have identified ROS producing enzymes such as Xanthine Oxidase that contribute to the pathobiology of radiation induced oxidative stress. 3) We have shown that inhibition of these enzymes can reduce oxidant stress, enhance vasoprotective nitric oxide production, and improve endothelial function and arterial compliance (a global measure for vascular health). 4) We have also demonstrated that administration of agonists that activate the endogenous Nrf2 ant-oxidant system can improve coronary endothelial function and blood flow. These finding have important implications not only for astronaut crew countermeasures but also have important implications for therapy and vascular protection of patients undergoing radiation therapy, or workers exposed to accidental radiation in the context of nuclear accident or in the event of terrorism. Finally, we have pioneered and perfected a laboratory technique used to measure endothelial function in vivo in mouse and rat coronary arteries. This has allowed us to test the effects of countermeasures on endothelial function in a vascular bed that is critically important.
Task Progress:	We have made significant progress addressing the potential role of radiation on coronary function. Resting coronary tone is significantly diminished in 5 Gy cohort ($p=0.0002$) as compared to sham radiated rats. Vasodilatory function is also attenuated. However, only difference in maximum vasodilatation is statistically significant ($p=0.0126$). Data demonstrates that radiation exposure decreases resting tone and vasorelaxation function. We have defined that it is Involvement of ROS that contributes significantly to the coronary endothelial function. Superoxide dismutase mimetic, manganese (III) tetrakis (4-benzoic acid)porphyrin chloride (MnTBAP), is administered as a bolus at 1 ug/kg in irradiated mice. Immediately following infusion, coronary flow velocity increases to levels observed in sham radiated rats 0 Gy level ($p=0.0042$). Data thus support that radiation exposure induces injury through increasing ROS levels. Data further support that scavenging ROS can reverse radiation-induced vascular damage. We are currently exploring ways of activating the antioxidant defense pathway, specifically antioxidant enzyme hemeoxygenase-1 (HO-1). We seek to determine if pre-activation of antioxidant defense mechanisms by the phytochemical sulforophane can defend against radiation injury in the coronary circulation. We have begun experiments to determine the role of antioxidant defense pathway activation as intervention against radiation injury. Sulforaphane, a NRF2 activator was injected at 5mg/kg intraperitoneally 48 and 24 hours prior to radiation exposure. Coronary flow velocity in sulforaphane-treated rats is higher than untreated rats, but change is not significant. Sulforaphane dose may not have produced a dramatic enough activation of the antioxidant defense pathway. Nevertheless, data supports that pharmacological activation the antioxidant defense pathway is a potential intervention method for radiation-associated coronary injury. We have initiated a pilot proposal recently funded by the NSBRI. We hypothesize that a targeted functional proteomic approach assessing oxidative protein modifications (OPTMs) will greatly enhance the sensitivity and specificity of detecting cardiovascular disease resulting from radiation induced injury. The proposed project will take advantage emerging technology that multiple covalent oxidative and reactive lipid and glycation modifications occur on plasma proteins in patients with pulmonary hypertension, systemic amyloid disease and in obesity. These results illustrate how blood components can be innocent passersby, modified in diseased tissue or in response to systemic metabolic changes. We will refine this approach by examining which proteins within the heart and vasculature are modified, determine which modifications occur in response to radiation, and identify a subset of these modified proteins in the plasma that demonstrate potential for use as tissue-specific biomarkers.

Bibliography Type:	Description: (Last Updated: 01/13/2014)
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